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AI Skills Improve Job Prospects: Causal Evidence from a Hiring Experiment

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The growing adoption of artificial intelligence (AI) technologies has heightened interest in the labor market value of AI related skills, yet causal evidence on their role in hiring decisions remains scarce. This study examines whether AI skills serve as a positive hiring signal and whether they can offset conventional disadvantages such as older age or lower formal education. We conducted an experimental survey with 1,725 recruiters from the United Kingdom, the United States and Germany. Using a paired conjoint design, recruiters evaluated hypothetical candidates represented by synthetically designed resumes. Across three occupations of graphic design, office assistance, and software engineering, AI skills significantly increase interview invitation probabilities by approximately 8 to 15 percentage points, compared with candidates without such skills. AI credentials, such as university or company backed skill certificates, only lead to a moderate increase in invitation probabilities compared with

self declaration of AI skills. AI skills also partially or fully offset disadvantages related to age and lower education, with effects strongest for office assistants, for whom formal AI certificates play a significant additional compensatory role. Effects are weaker for graphic designers, consistent with more skeptical recruiter attitudes toward AI in creative work. Finally, recruiters own background and AI usage significantly moderate these effects. Overall, the findings demonstrate that AI skills function as a powerful hiring signal and can mitigate traditional labor market disadvantages, with implications for workers skill acquisition strategies and firms recruitment practices.

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